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A novel Deep Learning architecture for lung cancer detection and diagnosis from Computed Tomography image analysis

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ABSTRACT

Timely identification of lung nodules, which are precursors to lung cancer, and their evaluation can significantly reduce the incidence rate. Computed Tomography (CT) is the primary technique used for lung cancer screening due to its high resolution. Identifying white, spherical shadows as lung nodules in CT images is essential for accurately detecting lung cancer. Convolutional Neural Network (CNN)-based methods have performed better than traditional techniques in various medical image applications. However, challenges still need to be addressed due to insufficient annotated datasets, significant intra-class variations, and substantial inter-class similarities, which hinder their practical use. Manually labeling the position of nodules on CT slices is critical for distinguishing between benign and malignant cases, but it is an unreliable and time-consuming process. Insufficient data and class imbalance are the primary factors that may result in overfitting and below-par performance. The paper presents a novel Deep Learning (DL) framework to detect and classify lung cancer in input CT images. It introduces a 3D-VNet architecture for accurate segmentation of pulmonary nodules and a 3D-ResNet architecture designed for their classification. The segmentation model achieves a Dice Similarity Coefficient (DSC) of 99.34% on the LUNA16 dataset while reducing false positives to 0.4%. The classification model shows performance metrics with accuracy, sensitivity, and specificity of 99.2%, 98.8%, and 99.6%, respectively. The 3D-VNet network outperforms previous segmentation methods by accurately calibrating lung nodules of various sizes and shapes with excellent robustness. The classification model's metrics show that the suggested method outperforms current approaches regarding accuracy, specificity, sensitivity and F1-Score.

1. Introduction

Globally, the leading factor for mortality rate is cancer, causing nearly 10 million fatalities in 2020 [1]. An estimated 25% of all global cancer-related fatalities are attributable to lung cancer. The uncontrolled division of damaged lung cells results in tissue tumors, eventually preventing organs from functioning properly. Lung nodules are the primary symptoms of lung cancer in the early stages. Lung cancer can be classified as Non-Small Cell Lung Cancer (NSCLC) or Small Cell Lung Cancer (SCLC). About 80%–85% of instances of lung cancer are caused by NSCLC. Compared to NSCLC, SCLC tends to grow and spread more rapidly, making about 10% to 15% of all lung cancer cases [2].

Machine Learning (ML) and Deep Learning (DL) have applications in the medical field to identify and interpret diseases, where DL can quickly and conveniently extract patterns and make predictions from large datasets as indicated by Santos et al. [3]. Convolutional Neural Networks (CNNs) are DL algorithms created primarily for processing and interpreting visual input, such as images and videos. Many CNNs used in the detection systems are 2D CNNs designed to work with 2D data, such as images, as shown in Fig. 1(a) [4]. They typically consist of CNNs that apply filters to the input image to extract features, followed by fully connected layers to classify the data. Xie et al. [5] and Dou et al. [6] state that some adaptations of 2D CNNs aim to incorporate multi-view planes or manually engineered 3D features to capture the 3D spatial information. 2D CNNs may not effectively leverage the 3D data. Various versions of 2D CNNs still cannot effectively capture the 3D spatial information in volumetric data because of the fundamental design of 2D network architecture. Ren et al. [7] and Dai et al. [8] suggest that numerous 3D CNNs and their modifications have been made to get around the drawbacks of 2D CNNs. Singh et al. [9], in his work, states that 3D CNNs consider the depth dimension and apply filters to the input volume to extract features from all dimensions. Fig. 1(b) [10] shows different slices of a chest Computed Tomography (CT).

Radiologists may now automatically detect lung nodules in chest CT scans using Computer-Aided Diagnostic (CAD) tools. Due to the

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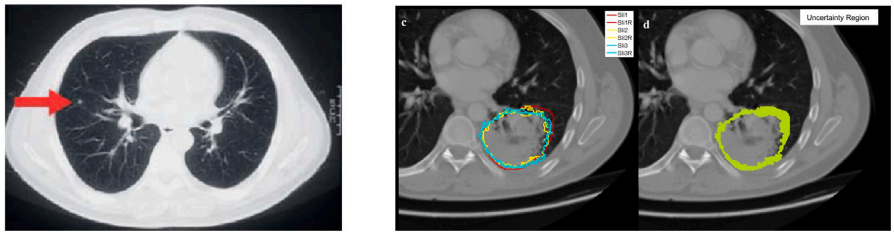


(a) 2D CT scan slice containing a small (5mm) early-stage lung cancer nodule [9]

(b) Volumetric CT-based segmentation of NSCLC using 3D-Slicer [10]

Fig. 1. CT slices for 2D and 3D CNNs respectively.

obscure characteristics of the lung cancer nodules, automatic diagnosis of lung cancer is now an arduous task. CAD systems are crucial for early cancer detection, reducing the evaluation time of radiologists and improving diagnostic accuracy. The CAD system for lung cancer detection encompasses various stages, including preprocessing techniques, segmentation, lung nodule detection, feature extraction, and lung nodule classification. The segmentation process is employed to emphasize the lung nodule in the CT scan, while classification is employed subsequently to determine whether the nodule is cancerous or non-cancerous. Although various methods for segmenting and categorizing lung cancer nodules are currently used in the literature, performance can be enhanced in both categories. Studies show that CT diagnostic systems backed by Artificial Intelligence (AI) may reliably identify lung cancer. With the potential for broader clinical integration, this breakthrough is highly significant for diagnosing lung cancer. ML and DL — as subbranches of AI, can speed up the identification of lung cancer, allowing researchers to examine many patients in less time. Although some CNN-based algorithms have offered cutting-edge performances for medical image segmentation applications, these systems still face some difficulties.

Applying DL methods for lung cancer detection can potentially increase the precision and speed of diagnosis. But for these methods to work, several obstacles must be addressed. The requirement for a significant amount of high-quality data is a primary limitation. Providing high-quality assurance, precise segmentation, and suitable annotations for data collection becomes difficult. Even with the introduction of automated techniques for such datasets, the outcomes still need to be more precise. The complexity of the images with several dimensions makes things more challenging because annotating an image may be tedious and time-consuming, especially for images with multiple slices. Therefore, a crucial need is for standardized data meticulously selected by qualified professionals. Expert knowledge and considerable time are needed to annotate medical images to identify lung cancer. Labeled data is necessary to ensure the creation and assessment of DL models, resulting in less-than-ideal performance. DL can automatically annotate data utilizing a large dataset of annotations. A better modal analysis can be achieved by designing CAD systems so that they can learn from both clinical records and medical images. Dodia et al. [11] states that data imbalance remains a significant concern when developing a real-time model. This has been found in cancerous and non-cancerous nodules in CT scans and when classifying the stages of cancerous nodules.

Issues like lack of high-quality data, image complexity, annotated data, and imbalanced datasets make it more difficult for neural networks to learn how to detect and diagnose lung cancer. Therefore, it is essential to create reliable DL models that concurrently address the challenges. To tackle the issues mentioned, this paper proposes performing lung nodule detection in 2 phases — the initial phase of lung cancer nodule segmentation followed by the nodule classification phase. CT scans produce high-resolution images of the thoracic region, which can result in large image files. The whole thoracic region, including the chest, heart, lungs, and ancillary structures, is captured in detail in these images. Although this level of distinction is helpful for diagnosis, it can also make image processing and analysis challenging. Early-stage detection frequently necessitates a detailed inspection of specific regions within the thoracic region, such as small masses in the lungs. These Regions of Interest (ROI) are relatively small compared to the whole CT image.

A smaller ROI for analysis is a crucial downside because the raw CT scan pictures of the thoracic region are often larger. For the proposed work, the actual input CT slice is of the size 512 × 512. Still, after preprocessing, it is divided into multiple patches of size 16 × 96 × 96, which is basically ($depth \times x - axis \times y - axis$), the chosen size to be fed in the input model for lung nodule segmentation, i.e., Volumetric CNN (3D-VNet). Later, annotations provided by radiologists in the dataset are used for the classification process, and images are further cropped to size 16 × 32 × 32 using lung nodule centers — the nodules and non-nodules generated as an output form an imbalanced dataset. A balanced approach to lung nodule classification is devised to avoid this issue. The model used to classify nodules as cancerous or non-cancerous is the 3D-Residual Neural Network (3D-ResNet). The segmentation model resulted in a Dice Similarity Coefficient (DSC) of 99.34%, where False Positives (FP) were reduced to an average of 0.4%.

The main contributions of this study are summarized below:

- A segmentation model, 3D-VNet is suggested to perform nodule region segmentation from a given CT image. This model improves the performance of the existing architecture by using DSC as a comparison metric.
- An algorithm that effectively balances the candidate lung nodules generated after the segmentation process.
- 3D-ResNet, a lung nodule classification model that improves existing state-of-art accuracy, sensitivity, and specificity, thereby reducing False Positive Rates (FPR).

The remainder of the paper is divided into the following sections: Section 2 presents an overview of related work in this domain. Section 3 describes the dataset utilized in this study. The proposed architecture is outlined in Section 4. The evaluation parameters for the working model are discussed in Section 5. Section 6 presents a comparative analysis of the results. Finally, Section 7 provides the conclusion by summarizing the key findings and gives direction for the future scope of the work.

2. Related-work

Multiple steps are involved in detecting and identifying lung cancer using DL architectures, each with unique tasks and objectives. These stages are broken down as follows: Image preprocessing is the initial phase, where the primary aim is to emphasize the ROI within the images from the lung CT scan. Intensity normalization, which provides uniform image intensity levels, and truncation of image values, which can aid in removing noise or unimportant information from the images, are some of the tasks involved. The subsequent step is lung nodule segmentation once the images have been pre-processed. This stage seeks to precisely define and hone the boundaries of nodule candidates within the images. This stage is significant since precise segmentation is necessary for the ensuing analysis. The emphasis turns to feature extraction and feature selection after lung nodule segmentation. Size, shape, texture, and intensity attributes are just a few of the features of the segmented nodules that are extracted at this stage. These features collect crucial data that can aid in determining whether a nodule is malignant. Additionally, feature selection is done to optimize computational efficiency and reduce the feature space's dimensionality to enhance the model's performance. The next stage is the classification of potential lung nodules using the extracted and chosen attributes. This process aims to increase the precision of lung cancer diagnosis and identification. DL architectures, renowned for automatically learning pertinent information, are frequently suitable for this task. This section extensively reviews numerous cutting-edge techniques developed for lung cancer diagnosis.

Initially, lung cancer detection was accomplished using ML models. These studies investigated various feature extraction and categorization methods. Techniques like Support Vector Machines (SVMs), decision trees, and others were utilized to categorize lung nodules as benign or cancerous. For example, Zhu et al. [12], in his studies based on several radiomic characteristics gleaned from CT scans, used an SVM to categorize lung nodules as benign or cancerous. The study achieved a considerable accuracy of 90.3% in detecting malignant nodules. The features and image processing methods used in traditional CAD systems are hand-crafted as described by Sluimer et al. [13]. However, conventional systems have several drawbacks. First of all, hand-selected feature extraction from images is not generalizable. This does not guarantee consistency in the feature selection. Second, there is redundancy in features following feature extraction. For more outstanding performance, many CAD systems employ several features; nonetheless, this ups the space complexity, rendering features superfluous.

A significant development in medical image analysis is the advent of DL-CAD systems. These systems address several issues with conventional CAD systems using DL methods, particularly CNNs and other deep neural network architectures. Medical imaging researchers use deep models since they can automatically learn features. As a result, DL models have become the prevailing trend in lung cancer diagnosis. Due to the power to automatically identify relevant characteristics from data, both 2D and 3D techniques utilizing DL architectures have become more favored. Compared to earlier ML techniques, these models have demonstrated enhanced accuracy. Different strategies have been explored in the context of 2D DL architectures. One illustration is a multi-view 2D CNN model, which accepts as input 2D slices from various angles of a CT scan. This method was employed in studies showcased by Roth et al. [14], which had a sensitivity of about 81%, and by Shah et al. [15], which had an accuracy of about 95%. This is a substantial improvement over earlier models.

Patch-based CNN models are another of the distinct 2D DL strategies used for lung cancer analysis. Small image patches are extracted from CT scans using these models, which categorize them as nodules or non-nodules. This approach has been used by Zhang et al. [16] and has shown an accuracy of 86.84% on nodule classification using only nodule patches. Studies have demonstrated the competence of 2D-DL models in improving the detection and diagnosis of lung cancer. Despite its benefits, 2D DL models might not be as good at finding small or precisely localizing nodules within the lung as 3D models. This limitation stems from the fact that 2D models cannot completely represent the volumetric data found in 3D scans.

A significant advancement in lung cancer detection was seen owing to the development of 3D CNN models. The complete CT scan volume is accepted as input by these models. The 3D method enables the model to capture the volumetric data in the images, which can be essential for identifying and classifying lung nodules. A 3D CNN was trained to recognize and categorize lung nodules by Anthimopoulos et al. [17]. Compared to earlier approaches, this strategy produced more precision and sensitivity results. Similarly, Tyagi and Talbar [18] used a 3D CNN to detect lung nodules and predict whether they will be malignant. When trained on the LUNA16 dataset and tested on a local dataset, the model obtained a DSC of 80.74%. The DSC is a statistic frequently used to evaluate how closely expected and actual segmentations resemble one another.

Bhatia et al. [19] presented a lung cancer detection method using deep residual learning. The researchers employed UNet and ResNet models to extract features indicative of cancer vulnerability from CT scans. To predict carcinogenicity, they utilized classifiers like Random Forest and XG-Boost. However, its accuracy in generating predictions constrained the model's performance. A DL algorithm that uses a CT image as a collection of states and calculates if a malignant nodule is present was proposed by Ali et al. [20]. They used a Reinforcement Learning algorithm that performs better with time and more data. The flaw was that the model was overfitting, and the authors suggested solving it with more data.

Another approach involves leveraging a combination of 2D and 3D CNNs. This analyzes both individual slices and the entire 3D volume of a CT scan. This hybrid approach tries to capitalize on the advantages of both 2D and 3D modeling methods. Yang et al. [21] suggested a hybrid 2D-3D CNN for lung nodule identification. Comparing this hybrid model to several other cutting-edge models, it showed higher performance, outperforming several further advanced models.

Ozdemir et al. [22] presented a novel Computer-Aided Detection (CADe) and diagnostic method designed for lung cancer screening using low-dose CT images. The system relies solely on 3D CNNs and excels in lung nodule detection and malignant growth classification tasks in the LUNA16 and Kaggle Data Science Bowl challenges. The authors have taken into account the interconnection between detection and diagnosis constituents. Leveraging this connection has enabled a comprehensive system with improved and more reliable performance, eliminating the necessity for an FP reduction stage in nodule detection.

Selvapandian et al. [23] introduced a lung nodule detection system developed using a Generative Adversarial Network (GAN) with the Sine Cosine SailFish (SCSF) algorithm. The SCSF algorithm enhances detection precision and effectiveness by merging the sine cosine algorithm alongside the SailFish optimizer. The discriminator's loss function is paramount in detecting affected areas precisely. The system achieves an accuracy of 94.74% for the first-level classification and 95.64% for the second-level classification on the Lung Image Database Consortium (LIDC-IDRI) dataset.

Chandrasekar et al. [24] showcased an approach called Multivariate Ruzicka Regressed eXtreme Gradient Boosting Data Classification (MR-RXGBDC) along with Service-Oriented Architecture (SOA) to improve the efficiency of lung cancer prediction in big data analytics by reducing prediction time and enhancing accuracy. The research enhanced the accuracy and efficiency of lung cancer prediction and analysis by including a diverse group of datasets such as the Lung Cancer dataset, the Air Quality-Lung Cancer Dataset, the Thoracic Surgery Dataset, and the LUNA16 Lung Cancer Dataset. For lung cancer detection, the MRRXGBDC method outperforms alternatives by 9% in prediction accuracy, 50% in FP reduction, and 11% in prediction time on the LUNA16 Lung Cancer dataset.

Costly annotations of CT images by domain specialists contribute to a need for labels and consistency between the patient's pathology evaluation and the observed nodule malignancy. Shen et al. [25] tackles these problems by presenting WS-LungNet, a two-phase weakly-supervised network designed for detecting and diagnosing lung cancer using CT scans. The network utilizes a Cross-Nodule Attention CADx to assess malignancy at the patient level and a semi-supervised CADe to segment pulmonary nodules using unlabeled data to mitigate label scarcity and inconsistency. The LIDC-IDRI dataset was used for training and testing through 10-fold cross-validation. The Competition Performance Metric (CPM) for the Semi-CADe (50%) model was 82.99%, and the accuracy and Area Under the Curve (AUC) for the Cross-Nodule

Attention CADx (CNA-CADx) component of WS-LungNet in assessing malignancy at the patient level were 85.72% and 88.63%, respectively.

Gautam et al. [26] presented a novel collection of DL models that integrates ensemble learning with deep transfer learning, using three sophisticated CNNs, i.e., ResNet152, DenseNet169, and EfficientNetB7, to boost the precision of lung nodule severity classification. An approach of assigning weights to each base model was devised to enhance the ensemble method's performance. The approach was evaluated extensively on the LIDC-IDRI CT scan dataset, achieving an accuracy of 97.23%. The authors affirm that the high accuracy is attributed to the innovative weight optimization strategy, which significantly decreased false negatives and increased sensitivity to 98.6%.

Heidari et al. [27] use DL algorithms to help identify the locality of malignant tumors. The challenge of analyzing data from multiple sources is addressed by combining Federated Learning (FL), Blockchain technology, and DL algorithms while maintaining patient privacy and trustworthiness during data sharing. To ensure data integrity and protection, the FBCLC-Rad framework, which integrates blockchain and edge technologies with lung cancer detection methods, is used. The CapsNets approach for classifying lung cancer in local mode is also used. Cancer Imaging Archive (CIA), LUNA16, Kaggle Data Science Bowl (KDSB) and the local dataset were used to train and assess the methods mentioned, obtaining an accuracy of 99.69% on the CIA and the local dataset.

By employing a region-based active contour model and YOLOv4, Dlamini et al. [28] create a fully automated system capable of identifying, delineating, and precisely reconstructing NSCLC tumors in three dimensions. The framework integrates detection and volumetric rendering techniques to analyze tumors in NSCLC. It involves image enhancement, augmentation, labeling, and localization in the detection phase, image filtering, tumor removal, region-based active contour and 3D reconstruction in the volumetric rendering phase. When evaluated on the LIDC-IDRI dataset during the testing phase, tumor segmentation, and extraction accuracy attained a DSC of 92.19%, which closely matched the ground truth.

A unique DL-based evolutionary technique termed GUNet3++ is proposed by Ardimento et al. [29], characterized by multi-scale skip connections that allow the network to gather information from contextual data over distinct areas of increasing sizes at numerous scales. The methodology applies a genetic algorithm within the GUNet3++ framework, an enhanced version of the generic UNet-based architecture, to identify the best architecture variation for segmentation. Genetic programming uses direct coding schemes and hyperparameter optimization to aid this method. A DSC of 0.978 between the predicted and reference ROIs is observed when the lesion possesses an irregular and asymmetric border and lacks calcification. This was performed on the Nodules Oracle dataset constructed using images from the LIDC-IDRI database with identical format and resolution.

DL Capsule Neural Network (CapsNet) presents an adequate replacement for CNNs to improve model comprehension of spatial hierarchies. Bushara et al. [30] propose a CapsNet architecture, Visual Geometry Group-Capsule Network (VGG-CapsNet), which utilizes CT images to classify lung cancer. VGG-CapsNet surpasses a single basic capsule network or an ensemble of CNN capsule networks, as indicated by its 0.980 AUC and 98.61% accuracy for the LIDC-IDRI datasets.

Rani et al. [31] presents a decision support system for classifying lung cancer. This system has four phases. The first step is to erase noise from the input image and enhance the contrast of the aberrant area in the image using Hybrid CLAHE combined with the Unsharp Masking Technique. The next step enhanced the quality of the segmented images and utilized the Radon Transform Based Improved Single Seeded Region Growing segmentation technique to detect the lung lesion accurately. In the third stage, the Gray Level Run Length Matrix and Gray Level Co-occurrence Matrix features were extracted from the segmented ROI image. The Advanced Marine Predator algorithm with SVM (AMPWSVM) Classifier is utilized to classify malignant and benign lung cancer images. An accuracy of 93.3% is attained using the AMPWSVM classifier on the LIDC-IDRI database, while a 90% accuracy is reached on the In-house clinical dataset.

Incorporating a combination of a CNN and a CapsNet, Bushara et al. [32] propose LCD-CapsNet, a DL framework to classify and identify lung cancer using CT imaging. This methodology effectively handles extensive quantities of data while maintaining spatial dependability. The results show that LCD-CapsNet surpasses the performance of CapsNet, achieving metrics: precision at 95%, recall at 94.5%, F1-Score at 94.5%, specificity at 99.07%, AUC at 0.989, and overall accuracy of 94% in classifying benign and malignant lung data on the LIDC-IDRI dataset.

The Bat Deer Hunting Optimization Algorithm-based Deep CNN (BDHOA-based DCNN) proposed by Navaneethakrishnan et al. [33] is an approach developed to mitigate the precision problems that arise during lung cancer detection in CT images. The BDHOA algorithm incorporates the Bat Algorithm (BA) and the Deer Hunting Optimization Algorithm (DHOA), where the lung lobe and nodule region are extracted from the lung image to perform the detection and classification of lung cancer. Validation and testing are conducted using the annotated thoracic CT scans from the LIDC-IDRI dataset. The Deep CNN model, enhanced by BDHOA demonstrated good performance in lung cancer detection, achieving accuracy, sensitivity, and specificity rates of 92.43%, 94.21%, and 89.15%, respectively.

Precise segmentation of small lung nodules continues to be a formidable task in medical imagery due to their low contrast and difficulty differentiating them from extraneous structures and noise in an imaging modality. Akila Agnes et al. [34] introduce Wavelet U-Net++, combining the U-Net++ framework with wavelet pooling. By employing the Haar wavelet transform to reduce the pixel depth of feature maps in the encoder, the approach improves the ability to identify intricate image details. The method was assessed on the LIDC-IDRI dataset, and a mean DSC of 0.936 and a mean Intersection over Union (IoU) of 0.878 in segmentation were obtained. The authors demonstrate that integration of wavelet pooling with Tversky and CE loss substantially improves the network's capability to delineate small and irregularly shaped pulmonary nodules precisely.

By combining Hybrid Wavelet Partial Hadamard Transform (Hybrid WPHT) and Discrete Local Binary Pattern (DLBP) on CT images, Venkatesan et al. [35] developed a hybrid diagnostic framework for lung cancer. Initial preprocessing of lung cancer CT images with adaptive median filtering was performed. Subsequently, features were extracted from the preprocessed images utilizing DLBP and Hybrid WPHT. The feature selection procedure was refined using the Adaptive Harris-Hawk Optimization (AHHO) method. Utilizing an Optimal SVM (OSVM)-based classification and an Improved Weight-based Beetle Swarm (IW-BS) algorithm for parameter tuning classifier's performance was improved. The research uses two datasets: a benchmark lung database of 300 images divided into three classes (normal, benign, malignant), each containing 100 images, and the LUNA16 dataset containing 1018 CT scans from the National Cancer Institute. The method exhibited notable performance on the LUNA16 dataset, attaining an accuracy of 99.55%, precision and recall of 99.33% each, specificity of 99.66%, F-Score of 99.33%, the rapid processing time of 1.01 s and AUC of 99.5%.

The Texture and Radiomics Features based Severity Classification (TRFSC) method proposed by Gupta et al. [36] utilizes 2D texture analysis and radiomics to extract crucial information from data pools. This method analyzes thoracic CT scans from the LIDC-IDRI dataset. The analysis employed optimal classification models constructed from 71 derived features from each patient in the LIDC-IDRI dataset, utilizing classifiers including SVM, Linear Discriminant Analysis, linear regression, k-Nearest Neighbours, Bayes, and boosted trees. The study's results demonstrated that the SVM classifier outperformed alternative classifiers, attaining an accuracy of 91.3%, specificity of 92%, sensitivity of 90%, and AUC of 96%. The ability to selectively apply different

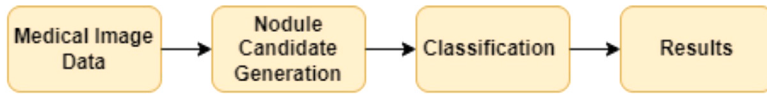


Fig. 2. Sequential process for identifying lung cancer using CNN architecture.

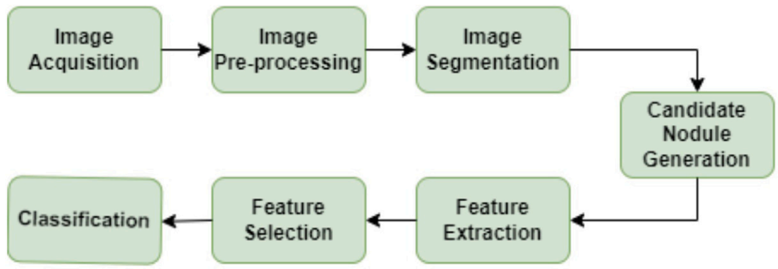


Fig. 3. Sequential process for identifying lung cancer using DL Architecture.

parametric values during feature extraction and class discrimination accurately underscores the approach to differentiating benign from malignant lung nodules.

These studies acknowledge the promising potential of ML and DL models for enhancing the identification of lung cancer with greater precision and effectiveness. However, additional study is required to validate these models on larger, more varied datasets and assess how well they work in clinical settings. This paper discusses 3D architectures for nodule extraction using volumetric segmentation and classification.

3. Dataset

The LUNA16 dataset, a subset of the larger LIDC-IDRI, is an essential resource for lung cancer research. It contains 1018 examples of thoracic CT scans. In contrast to LIDC-IDRI, LUNA16 exclusively comprises scans with a slice thickness below 2.5 mm and encompasses a total of 888 cases, featuring a collective count of 1186 lung nodules. With LUNA16 having a mean nodule size of 8.3 mm and LIDC-IDRI having a mean nodule size of 12.8, there is a considerable divergence in the nodule size distribution (Li and Fan [37]). The dataset comprises 888 chest CT images, each measuring 512 × 512 pixels. Each 512 × 512 pixel image is broken into ten subgroups using the MetaImage format (.mhd and .raw). The annotations.csv file contains annotations that classify lesions as nodules or non-nodules based on size. Lesions are categorized according to their size ($nodule \leq 3$ mm, and $nodules \geq 3$ mm) [38]. The 3D-VNet model for lung cancer detection and the 3D-ResNet model for lung cancer identification used in this study have been trained using the LUNA16 dataset.

4. Proposed methodology

Fig. 2 depicts a general technique for identifying lung cancer based on conventional approaches. In contrast, Fig. 3 illustrates a generalized detection system based on DL architecture [39]. The subsequent sub-sections provide more detailed explanations of each step involved.

The first stage in creating a CAD model requires collecting images from various imaging methods by systematically scanning the patient to capture images. Multiple modalities, such as X-rays, CT scans, and Magnetic Resonance Imaging (MRI), can be employed. For instance, the chest/thoracic region imaging techniques, such as chest X-ray, CT scans, and MRI, are depicted in Fig. 4(a), Fig. 4(b), and Fig. 4(c), respectively.

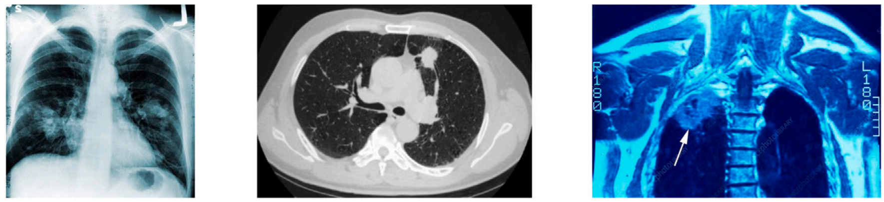
The lung nodule detection and classification process is divided into three main stages. The proposed model's schematic diagram is shown in Fig. 5.

1. **Preprocessing Stage:** The stage takes the input CT scan of size 512 × 512 pixels ($x\_axis \times y\_axis$) from the LUNA16 dataset. The third dimension of the image (z_axis) is the number of slices in each CT scan. Fig. 6 shows a 3D visualization of the input image across the 3 axes. Table 1 lists the minimum and the maximum values that the z_axis can take in each of the subsets of the dataset. Image processing techniques are applied to the input images, such as resizing, intensity normalization, and truncation of image values. The masks for each slice are computed, and the images and the mask are transformed into numpy arrays. Subsequently, both are divided into 16 patches, each measuring 96 × 96, used as input for the proposed segmentation model, i.e., 3D-VNet.
2. **Lung Nodule Extraction Stage:** This stage focuses on extracting lung nodules following segmentation. The model identifies nodules associated with the images and extracts cubes around the nodules, which are 3D-CT cubes. These cubes symbolize areas of interest in the three-dimensional space. The model saves the nodules based on class labels: 0 for non-nodules or 1 for nodules.
3. **Classification Stage:** The classified data from the previous stage is highly imbalanced. A balanced dataset of nodules and non-nodules is created to reduce the data skewness. Then, classification is done using fused features obtained from the 3D-ResNet model and custom-made features.

The following sections provide a detailed explanation of these phases. Section 4.1 describes how to process CT scans using a 3D model to gather volumetric data and how features and slice expansion affect the number of slices produced. It also introduces a slicing and expansion algorithm. Section 4.2 explains the significance of ROI segmentation in CT scans and presents the utilization of a 3D CNN model for this purpose. The post-segmentation process, candidate nodule detection, categorization, and the unequal distribution of candidate nodules are described in Section 4.3. This subsection also describes the strategy used to solve the problem of an unbalanced dataset and the procedures for producing a balanced dataset for use in the training and testing of the DL model. Section 4.4 outlines the architecture of the proposed 3D-ResNet classification model.

4.1. Image preprocessing

The LUNA16 dataset comprises 888 lung CT scans. The 3D model used in this work requires processing numerous slices from a single CT scan to obtain volumetric data about the lung under consideration. Unlike 2D models, which deal with individual slices, a 3D model processes several slices from a single CT scan to produce a thorough and accurate three-dimensional depiction of the lung. This illustration offers a more



(a) Chest X-ray (b) CT scan (c) MRI scan

Fig. 4. Different imaging modalities for the lung.

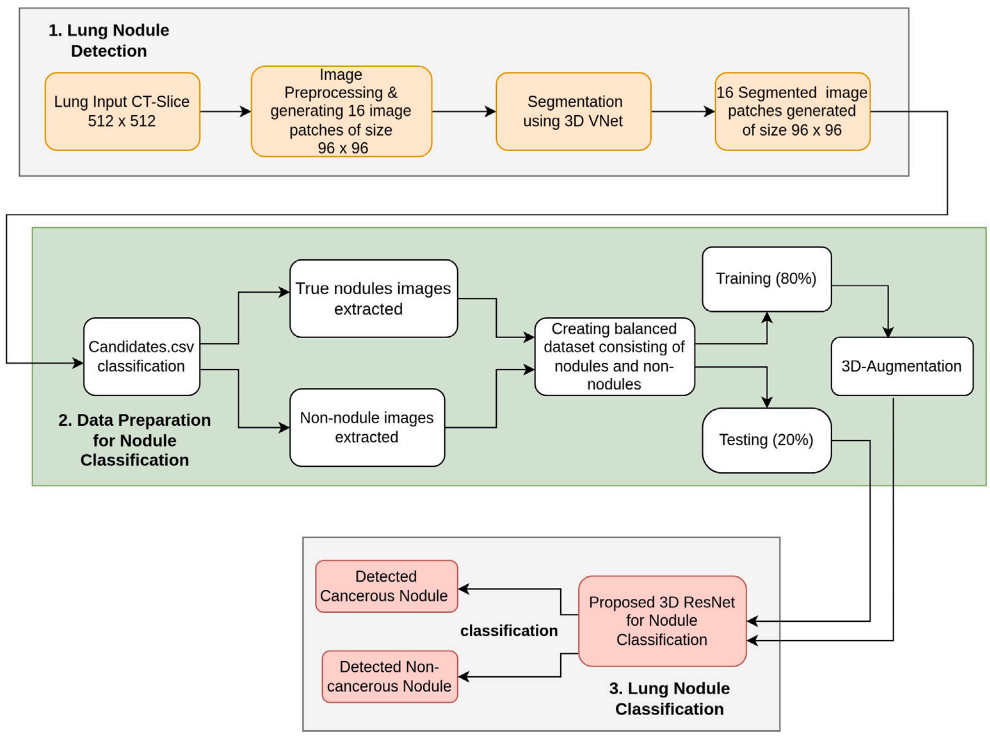


Fig. 5. Schematic diagram for proposed model.

Table 1
Range of slice thickness across LUNA16 dataset subsets.

Subset Number	Number of images in the subset	Lowest number of slices in an image (minimum value)	Highest number of slices in an image (maximum value)
0	89	109	733
1	89	103	636
2	89	95	682
3	89	107	764
4	89	109	564
5	89	109	633
6	89	117	590
7	89	117	522
8	88	111	580
9	88	113	567

exhaustive and detailed awareness of the structure of the lung and can facilitate analysis, visualization, and decision-making. The features of the original scan, such as the resolution and the field of view, along with the degree of slice expansion applied to generate additional slices from the original ones, influence how many slices are generated for a single scan. This procedure is described in algorithm 1, which shows how the slicing and expansion contribute to the total number of slices produced. Fig. 7 showcases how multiple slices are generated after processing a single CT scan. These CT slices are larger (512×512), and the region consisting of lung nodules in each slice is smaller. Instead of considering the entire slice at once for segmentation, this work adopts a patch-based approach. The patch-based segmentation approach divides the given slice into smaller overlapping or non-overlapping 16 patches of size 96 × 96 as shown in Fig. 8. The patch size of 96 × 96 pixels strikes a balance between capturing minute details and maintaining the context needed for accurate segmentation.

By analyzing smaller patches, the segmentation algorithm can capture fine-grained details and variations within the image, improving segmentation results. Large CT scans, such as the ones taken into account as input to the preprocessing stage, contain global information and sections beyond the intended ROI. Processing large images requires extensive computational resources depending on the complexity of the

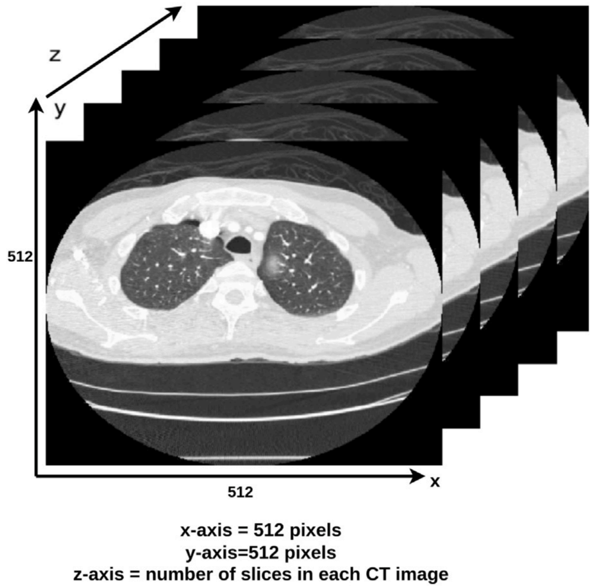


Fig. 6. 3D visualization of CT slices across x, y and z-axes.

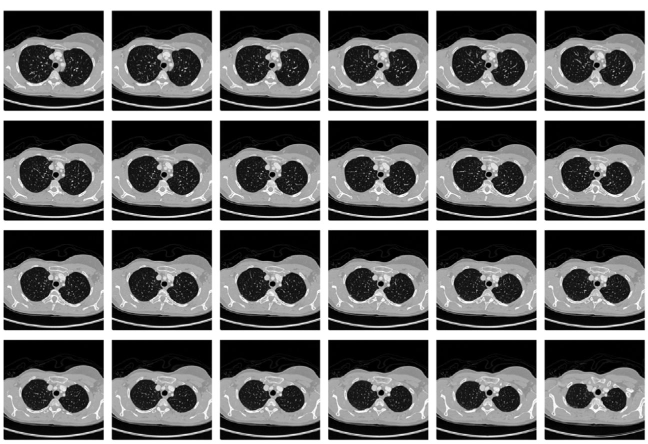


Fig. 7. CT slices generated from a single CT scan.

images, and it may take more time to process them. This is because the large images have contextual information with high granularity, which is beyond the scope of interest. In contrast, having smaller patches helps the segmentation algorithm focus on the intended ROI. Smaller patches can also be processed parallelly, speeding up the overall computation required. There is also a low possibility for the segmentation algorithm to miss out on critical details, which are the characteristics to perform segmentation of the lung nodules.

In Algorithm 1, $stpos$ represents the start position, and $endpos$ represents the end position. These two variables represent the range of image depths. $imageZ$ represents the number of slices in the $segLungImg$, i.e., the CT image to be analyzed. The algorithm uses the $getRangImageDepth(segLungImg)$ function that returns the $stpos$ and $endpos$ values, which indicates the range of valid slices within the $segLungImg$. This function assesses the content of $segLungImg$ and identifies where the relevant data begins and ends along the depth dimension. The algorithm modifies two arrays, $srcImg$ and $segLungImg$ arrays, by performing slicing and expansion as per conditions. The slicing operation is performed along the first dimension of the array. This implies that you are only extracting certain portions of the depth-based data. After the slicing and expansion operations, the resulting sliced arrays are assigned back to $srcImg$ and $segLungImg$, respectively.

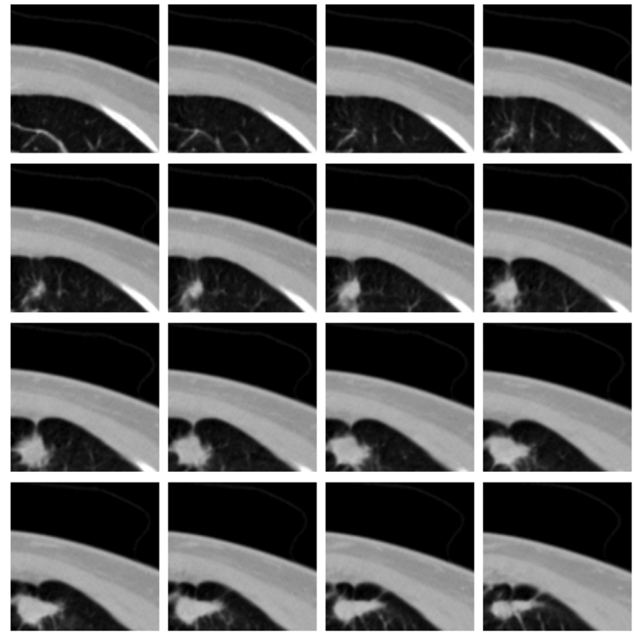


Fig. 8. 16 image patches of size 96 × 96 derived from a slice.

This algorithm is essential for focusing on a specific ROI within the image data.

Algorithm 1 Slicing and Expanding Image Data.

Require:
- $segLungImg$: 3D array representing segmented Lung Image
- $srcImg$: 3D array representing source Image
- $expandSlice$: a non-negative integer indicating the expansion amount.

Ensure: Modified $srcImg$ and $segLungImg$ based on slicing and expansion

$stpos, endpos \leftarrow getRangImageDepth(segLungImg)$
if $stpos == endpos$ **then**
  continue
end if
$imageZ \leftarrow segLungImg$
$stpos \leftarrow stpos - expandSlice$
$endpos \leftarrow endpos + expandSlice$
if $stpos < 0$ **then**
  $stpos \leftarrow 0$
end if
if $endpos > imageZ$ **then**
  $endpos \leftarrow imageZ$
end if
$srcImg \leftarrow srcImg[stpos : endpos, :, :]$
$segLungImg \leftarrow segLungImg[stpos : endpos, :, :]$

The lung nodule and surrounding neighborhood area have significantly different radio densities. Thus, we require an efficient preprocessing of these CT slices to highlight the lung nodule location distinct from the rest of the search area, making it easy to locate the ROI. The image preprocessing pipeline includes a series of operations, such as

- Image resizing: Varying the size of the images.
- Noise reduction: Lowering noise in the images.
- Thresholding: Generating binary images based on a threshold value.
- Image enhancement: Enhancing the features and quality of images.
- Normalization: Regulating image values to a standard scale.

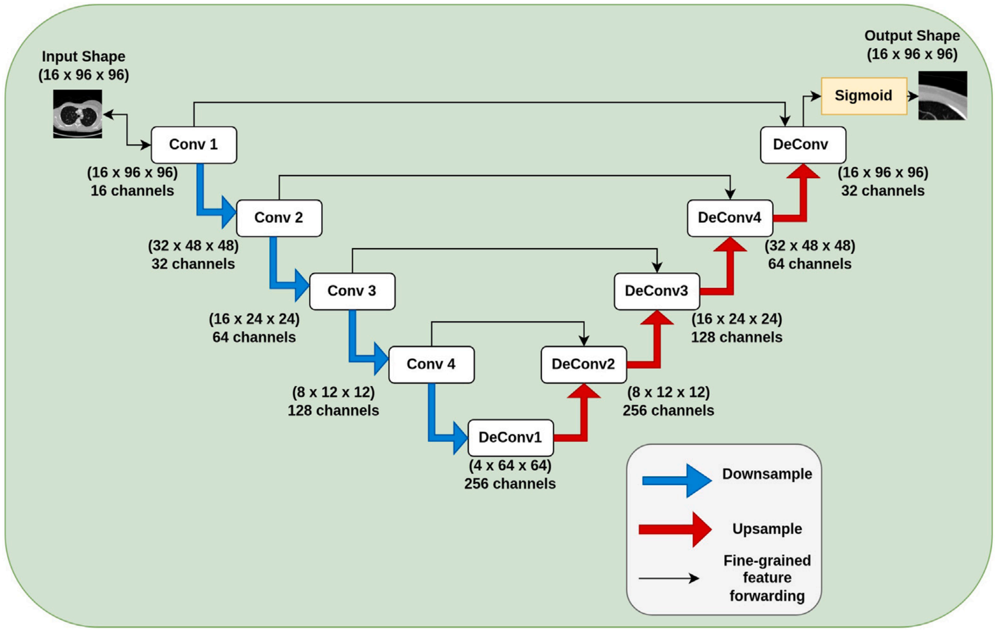


Fig. 9. The proposed 3D-VNet architecture.

• Resampling: Altering the resolution of the images.

The preprocessing pipeline is intended to improve the effectiveness and efficiency of lung nodule detection. The pipeline aims to highlight lung nodules and their surrounding areas while mitigating unwanted variations and noise in the CT scan images. It accomplishes this by standardizing image size, reducing noise, creating binary representations, enhancing image quality, normalizing pixel values, and resampling for constant resolution. These image preprocessing methods guarantee that the CT scan images of the lungs have been processed for lung nodule detection.

In this work, preprocessing is also conducted by quantifying the distinct radio densities. These densities are measured using Hounsfield Units (HU). Different tissues and structures in the body have distinct HU values. The ROI of the slices is limited to the range of −1000 to 400 HU to eliminate variations caused by image resampling and bone densities. The images are normalized after HU-based quantification and range restriction. The process of normalization involves converting the intensity values of the images to a standard scale. Accordingly, in this work, the normalized images are adjusted to have a mean voxel value of 0 and a variance of 1. This standardizes the intensity values of the images. Lastly, thresholding is performed using grayscale conversion on the normalized images, after which they are used as input for the segmentation process to detect lung nodules. Grayscale conversion refers to changing an image's original color format to grayscale. The images that have been processed and thresholded are then fed into the segmentation step that seeks to identify lung nodules. With simplified grayscale information and standardized intensity values, the thresholded images are well-prepared for the segmentation algorithm to identify lung nodules.

4.2. 3D-VNet: Segmentation architecture

Although CT scans normally record a lot of information, the structures of interest, like lung nodules, are usually tiny and situated within the ROI. The complex composition of medical images poses challenges in effectively identifying and extracting the ROI. A segmentation process with suitable techniques isolates and distinguishes the ROI from the surrounding background in the CT scans. The ROI must be correctly identified and extracted, which can be challenging given the complicated structure of medical images. This study focuses on utilizing 3D representations of CT slices for segmentation. 3D representations capture the spatial relationships and context between slices, which can be valuable for accurate segmentation. This presents a demanding task due to the voluminous information involved. Therefore, a 3D CNN model tailored explicitly for segmenting volumetric images, namely 3D-VNet, is introduced. 3D-VNet, i.e., Volumetric CNN, is an extension of the 3D-UNet architecture. It captures multi-scale image features by leveraging skip connections and effectively segments volumetric data by directly processing 3D volumetric data. 3D-VNet effectively extracts lung nodules from a given CT image in our work. The input to this model is a CT image of size 16 × 512 × 512, i.e., $kernel\_size \times x\_axis \times y\_axis$. According to this input size, each CT image is represented as a 3D volume with 16 slices, and each slice has dimensions of 512 pixels along the x-axis and 512 pixels along the y-axis. The 3D-VNet architecture is shown in Fig. 9.

3D-VNet differs from UNet by utilizing 3D convolutions instead of 2D convolutions to analyze volumetric data, enabling the capture of spatial information in three dimensions. 3D convolutional layers, a symmetric architecture, residual connections, and loss functions are the fundamental components of a 3D-VNet. The network uses the 3D convolutional layers to process 3D volumetric data from the images directly. It is necessary to capture the spatial environment in three dimensions for accurate segmentation.

To process volumetric data, such as 3D medical scans or video sequences, 3D-VNet uses the encoder–decoder structure of the UNet. The terms "encoder" and "decoder" designate the elements in charge of the symmetric architecture's compression and decompression, respectively. V-Net, similar to U-Net, includes a contracting path to decrease resolution (downsampling) and an expansive way for increasing resolution (upsampling). The architecture depicted in Fig. 9 has two branches, each in a different color. The branch depicted in blue is associated with down-sampling. It is responsible for performing compression. The down-sampling technique, which is a compression type, decreases the input data's spatial dimensions. Convolutions with a stride of 2 are employed to achieve downsampling; this reduces the spatial dimensions while increasing the feature dimensions. The branch depicted in red represents the up-sampling operation. The up-sampling operation decompresses the signal, returning it to its original dimensions. The layers involved in upsampling, also known as deconvolution layers, enhance the spatial resolution of feature maps, resulting in better localization accuracy of the ROI in the segmentation map.

Compression and decompression are the two fundamental pathways of 3D-VNet architecture. The compression path is located on the left, while the decompression path is on the right. Data is compressed before transmission over the compression path. The architecture comprises five encoder blocks, each formed with convolutional layers, batch normalization, Rectified Linear Unit (ReLU) activation, and down-sampling operations. At each stage within an encoder block, convolution operations are performed. These operations extract hierarchical features and learn features from the data. The encoder blocks consist of five layers with filter sizes of $5 \times 5 \times 5$, $5 \times 5 \times 5$, $3 \times 3 \times 3$, $3 \times 3 \times 3$, and $1 \times 1 \times 1$. Each of these layers has a stride value of 1, which denotes the step size of the convolution operation.

The decoder component receives the encoded or compressed data and performs the decoding process. The decoder undergoes a decompression procedure using an up-sampling operation to reassemble the compressed data to its original dimensions. The Up-sampling operation aims to match the spatial dimensions of the corresponding skip connection feature maps from the encoder. A skip connection is a connection that enables data to be transmitted from one network layer to a subsequent layer. These skip connections are used by the architecture to merge feature maps from the downsampling path with those from the upsampling path. These connections help recover spatial information lost during downsampling, thus improving the accuracy of segmentation.

The up-sampling operation restores the dimensions of the feature maps to align with those from the encoder. Each decoder block integrates both low-level and high-level characteristics. The up-sampled feature maps are used with the encoder's skip connection feature maps to get this result. By utilizing the data provided at various network levels, this process produces segmentation maps that are more precise and detailed. The architecture includes five decoder blocks, each consisting of filter sizes of $1 \times 1 \times 1$, $3 \times 3 \times 3$, $3 \times 3 \times 3$, $5 \times 5 \times 5$, and $5 \times 5 \times 5$. Like the encoder blocks, all of the decoder blocks' layers have a stride of 1. The residual connections present in the architecture improve the gradient flow in the network throughout training. These connections enhance training stability by mitigating the vanishing gradient problem. A batch or group normalization is applied after each convolutional layer. This helps in stabilizing and expediting the training process. ReLU activation functions are applied after normalization to add non-linear elements to the model, allowing it to capture complex patterns for the ROI. The last layer employs a sigmoid activation function to produce the segmentation map suitable for binary segmentation mappings. A dropout mechanism with a rate of 0.5 is employed to prevent overfitting. During training, dropout sets a portion of the input units to zero at random. This technique promotes generalization by limiting the network's reliance on specific properties during training. The architecture of the proposed 3D-VNet for segmentation is outlined in the flow diagram Fig. 10(a).

Eq. (1) shows the deconvolution operation 'De' with various parameters like input data as 'x', kernel or filter as 'k', feature map size as 'f', and stride as 's'

$$De(x, k, f, s) = ReLu(De(x, k, f, s)) \tag{1}$$

As depicted in the equation above, each output of a deconvolution layer is first processed by a ReLU activation function before being fed as input to the subsequent deconvolution layer. The deconvolution result is processed by applying the ReLU activation function on the output. The ReLU activation function, as seen in neural network structures, brings non-linearity to the output.

Eq. (2) shows the convolution operation, which takes the following parameters '$Conv$' with input data 'x', kernel or filter 'k', and stride 's'.

$$Conv(x, k, s) = sigmoid(Conv(x, k, s)) \tag{2}$$

The equation above demonstrates that each convolution layer's output is subjected to a sigmoid activation function before being used as the input for the following convolution layer. The output of the equation is sent through a sigmoid activation function once the convolution is computed. The sigmoid activation function restricts its output to 0 and 1, making it appropriate for binary categorization and probability calculations. Both equations highlight that the result is passed through a ReLU activation function after each deconvolution operation. Similarly, each convolution operation is followed by the result being processed through a sigmoid activation function.

4.3. Data processing

After the segmentation process, we get segmented images of the size $16 \times 96 \times 96$. These images most likely show the relevant segmented areas of lung cancer nodules. To ensure accurate identification of whether they contain the actual lung cancer nodules, these images are further cropped to the region of size $16 \times 32 \times 32$. This step concentrates on the critical areas of the segmented images. The candidates.csv contains information about the nodule candidates supplied by the LUNA16 dataset. Along with the list of the nodule candidates, the candidates.csv file also includes the spacial coordinates of the nodules. The parameters consist of the following: seriesuid, coordX, coordY, coordZ, and class, where class denotes whether the candidate is a genuine nodule. The coordinates specified for each candidate nodule in candidates.csv serve as the center for the subsequent cropping operation. The cropping method involves creating a reduced volume concentrated around specific coordinates to emphasize the ROI. With this information, candidate detection techniques are used to compute the candidate locations.

There are a total of 5,51,065 candidate nodules in the dataset. These prospective nodules have the potential to cause lung cancer. The candidate nodules are then categorized as either true nodules (signifying locations that are lung cancer nodules) or non-nodules (illustrating regions that are not lung cancer nodules). The classification process produces a highly unbalanced distribution of candidates. It yields 1351 true nodules and 5,49,714 non-nodules. This imbalance means that the dataset contains significantly more examples of non-nodules than true nodules. This vast disparity can make training and classification difficult since models may be biased towards the dominant class. Due to the imbalanced distribution, the dataset becomes unsuitable as input for the training model 3D-ResNet. As a result, there is a considerable risk of misdiagnosis because the model may struggle to classify minority class correctly.

To counter the issue of class imbalance, this work proposes forming a balanced dataset. This balanced dataset would consist of all 1351 true-nodules from the true-nodule set and 1351 non-nodules randomly selected from the non-nodules set. We partition the balanced dataset using an 80:20 split to create training and testing sets. 80% of nodules and non-nodules are designated for training, with the remaining 20% reserved for testing. The training set consists of 2160 samples, with 1080 nodules and 1080 non-nodules, whereas the test set comprises 542 samples, with 271 nodules and 271 non-nodules. To further enhance the robustness of our model and expose it to a wider variety of data scenarios, we augment the training set. The DL model works best with larger datasets, so a 3D augmentation is applied to the training set. Augmentation creates new training samples by applying various transformations to the existing images. 3D augmentation includes techniques such as horizontal and vertical flips — images are flipped along the horizontal and vertical axes; rotation — different angles are used to rotate images; translation — translation by a factor ranging from 0 to 1; scaling — random scaling is applied with a probability factor, shearing, etc. The images also undergo transpose operation by swapping rows and columns. The affine transform augmentation is applied consistently with a translation ranging from 0 to 1 and random scaling with a 0.5 probability. To further augment the dataset, the images are horizontally and vertically flipped with a probability of 0.7. When all the augmentations are applied, a dataset of 86,400 nodules is

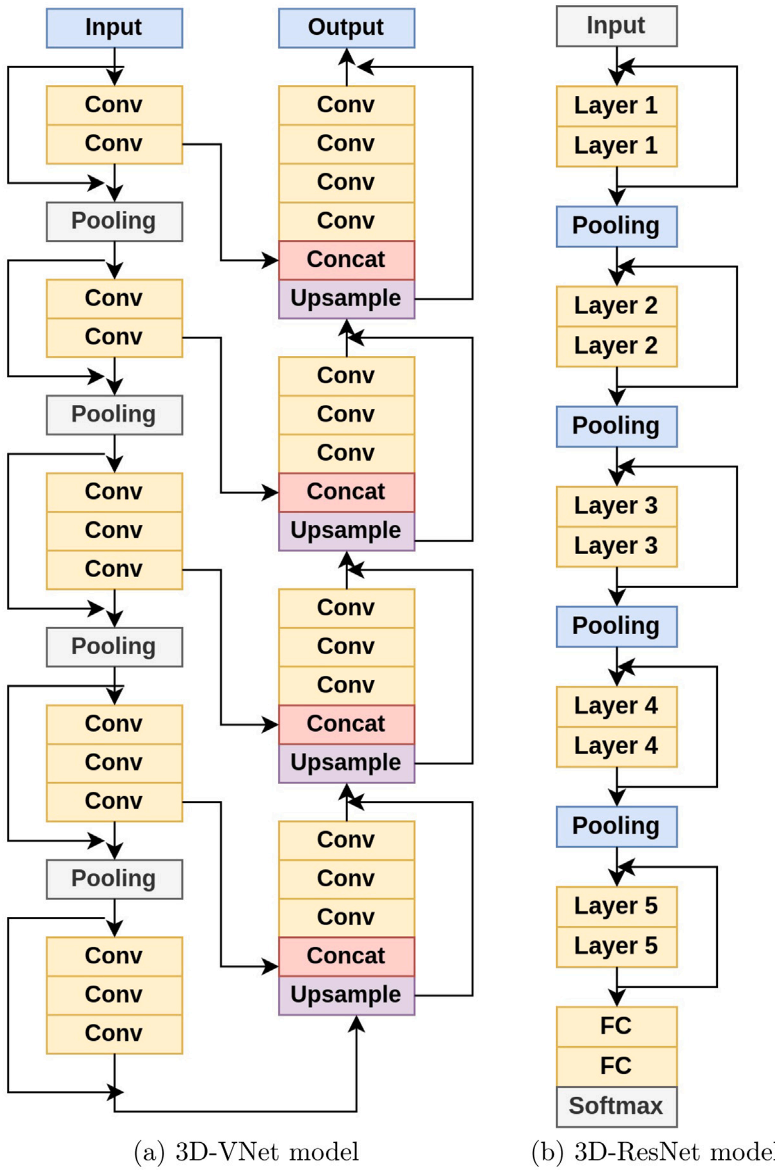


Fig. 10. Schematic flow diagram of architecture applied.

created, with 43,200 nodules each in the true nodules and non-nodules subsets. Augmentation is only performed on the training set. The test set remains at 20% of the original dataset size, guaranteeing that the model's performance is assessed on unaltered, real-world data.

4.4. 3D-ResNet: Classification architecture

The 3D-ResNet framework is designed for classifying 3D volumetric data. The schematic diagram for the proposed classification architecture 3D-ResNet is shown in Fig. 10(b). The output of this classifier would indicate whether or not a true nodule is cancerous. The input image to this classifier model is 16 × 32 × 32. This input image is passed through multiple layers of organized CNN structure. Utilizing 3D convolutional layers is essential for analyzing volumetric data as they allow the network to understand spatial hierarchies of characteristics within 3D input data. This capability facilitates the network's analysis of volumetric images by maintaining the spatial context throughout the image slices.

In the proposed architecture, Layers 0 and 1 employ a 3D convolution operation with 16 filters and a 3 × 3 × 3 kernel size. The output from Layer 0 is integrated with Layer 1 using a residual connection, facilitating the seamless flow of information through the network. To prevent overfitting, ReLU activation functions and dropout techniques are applied. Including non-linearities following each convolutional layer in the ReLU activation function builds the capability of the model to detect mappings between inputs and outputs. Non-linearity to the model helps to avoid the vanishing gradient issue. Dropout is incorporated after the activation function to reduce the likelihood of overfitting. This dropout randomly deactivates a section of the neurons during training, reducing overfitting. Each convolutional layer's output is normalized using a group normalization layer. Normalization is necessary to ensure a consistent distribution of activations, speed up convergence, and enhance the model's performance. Residual connections, a vital characteristic of ResNet architectures, allow the network to skip over one or more layers. These connections help address the vanishing gradient issue and improve the training of deeper networks by increasing the flow of gradients across the network.

Down-sampling is performed using max pooling, reducing the spatial dimensions of the data. Max pooling layers decrease the dimensions of feature maps through downsampling while preserving critical information. This strategy alleviates the computational workload on subsequent layers. Layer 2 consists of two convolutional layers, each comprising 32 filters, and these layers, like Layers 0 and 1, are also combined using residual connections. This pattern of combining layers through residual connections continues in subsequent layers until Layer 5. Layer 5's output is flattened and fed into fully connected layers. Global average pooling is applied to each feature map before the final fully connected layers to reduce it to a single value. In contrast to densely connected layers, sparsely connected layers effectively decrease the number of parameters, thus reducing overfitting. The network transitions from convolutional layers to fully connected layers to perform classification.

The First Fully Connected (FC1) layer comprises 512 units and applies ReLU activation and dropout. The global average pooling layer output is flattened and fed into fully connected layers. The network ends with a softmax activation function for binary-class classification, i.e., cancerous and non-cancerous nodules. The Xavier initialization method is used to initialize biases and weights. This extensive initialization approach guarantees uniform allocation of activations across the network. The architecture utilizes backpropagation training methods, with an accuracy metric for assessment and a classification-specific cost function (cross-entropy). Inference requires using previously stored weights and biases to classify fresh inputs.

The equations for the convolution, normalization, activation, and dropout operations are represented below:

$$conv = Conv3D(x, W) + B \tag{3}$$

Eq. (3) performs a 3D convolution operation on input 'x' using the weight tensor 'W' and adds the bias vector 'B' to the result. Eq. (4) applies a normalization layer to the convolution result '*conv*'. The specific normalization technique and equations depend on implementing the '*norm*' function. The parameters phase (ph), height (x), width (y), z as image dimension, type as normalization type, and scope (S) are passed to the function as arguments.

$$conv = norm(conv, ph, x, y, z = imageZ, type = group, S) \tag{4}$$

Eq. (5) applies the ReLU activation function to the convolution result '*conv*' and then performs dropout on the activated tensor. The ReLU function applies the Rectified Linear Unit activation, and the Dropout function applies dropout with a specified dropout rate.

$$conv = Dropout(ReLU(conv), drop) \tag{5}$$

To avoid overfitting, dropout randomly deactivates a portion of neurons during training. The 3D-ResNet architecture employed in this work leverages residual connections to enhance information flow within the network. It begins with input data processed through convolutional layers, down-sampling, and fully connected layers, ultimately producing the final output of lung nodules detected as cancerous or non-cancerous.

5. Performance evaluation

Performance evaluation of a model is essential to selecting the best approach and ensuring it generalizes effectively to new data. Standardization enables comparisons with existing models and benchmarks, facilitating model development and assessment advancements. Model validation and evaluation select the best architecture, hyperparameters, or training strategies. A Confusion Matrix is an essential tool for assessing the performance of classification models. It considers how well a system classifies data into actual and predicted categories. By comparing the model's predictions with the true labels, it summarizes classification accuracy. In the context of detecting lung cancer, four output parameters of the Confusion Matrix are defined as follows:

1. True Positives (TP): These are instances where the model accurately detects the presence of lung cancer, correctly classifying them as belonging to the target class, which are the positive cases.
2. True Negatives (TN): These are instances where the model correctly classifies individuals as not having lung cancer, accurately categorizing them as negative cases, which belong to the non-target class.
3. False Positives (FP): These instances represent incorrect classifications where the model falsely identifies individuals as having lung cancer despite them not belonging to the target class.
4. False Negatives (FN): These are cases in lung cancer detection where the model makes incorrect predictions by wrongly classifying individuals as not having lung cancer, even though they do belong to the target class.

The suggested model's performance is assessed using a variety of measures, some of which are outlined below:

- Accuracy: It determines the proportion of occurrences correctly categorized from the total number of cases in the dataset.

$$Accuracy = \frac{TP + TN}{TP + FP + TN + FN} \tag{6}$$

- DSC: Measures overlap between the predicted segmentation and ground truth masks.

$$DSC = \frac{2 * TP}{TP + FP + FN} \tag{7}$$

- Recall: It measures the proportion of lung cancer cases that the model correctly identifies.

$$Recall = \frac{TP}{TP + FN} \tag{8}$$

- FPR: When the FPR value is low, the model produces fewer false positives, which is suitable for classifying negative instances.

$$FPR = 1 - \frac{TP}{TP + FP + FN} \tag{9}$$

A lung cancer detection model is considered effective when it exhibits high DSC, recall, accuracy, and a low FPR. Furthermore, to evaluate its performance comprehensively, other metrics can be utilized as follows:

- Precision: It determines the proportion of positive instances correctly predicted as positive.
- F1 Score: A weighted average of recall and precision is used to get the F1 score.
- The FPR and recall are plotted on the ROC curve. The ROC curve's performance improves as it gets closer to the graph's top-left corner.
- AUC refers to the region under the ROC curve. A greater AUC value signifies superior model performance.

6. Results and analysis

3D-VNet and 3D-ResNet are advanced DL models used in medical imaging to precisely segment lung nodules from CT scans and categorize them as cancerous or non-cancerous. These models efficiently evaluate intricate structures in CT images by utilizing spatial hierarchies. Studies have shown that 3D models exhibit better performance than 2D models. Studies have resulted in the fact that the 3D models exhibit better performance when compared to the 2D models. This is attributed to the enhanced ability of 3D models to capture nodule information, surpassing that of 2D slices from CT scans. Including volumetric data in the 3D format significantly contributes to the overall improvement in system performance.

Table 2
Comparison of DSC of segmentation architectures on LUNA16 dataset.

Reference	Architecture	DSC%
Mohammed et al. [40]	VNet	80.0
Zhang et al. [41]	UNet	83.2
Bhattacharyya et al. [42]	Hierarchical attention UNet (HAUNet-3D)	83.3
Liu et al. [43]	Deep Neural Network VNet	88.3
Boubnovski et al. [44]	A multi-task learning VNet	94.0
Dodia et al. [11]	Regularized VNet and NCNet	95.0
Xiao et al. [45]	UNet and Res2Net	95.3
Bansal et al. [46]	Deep 3D Segmentation work (Deep3DSCan)	95.8
Sathish et al. [47]	2D CNN	98.4
Ours	**Proposed 3D-VNet Architecture**	**99.3**

Table 3
Confusion matrix for Case 1.

Trial no.	TN	FP	FN	TP
1	239	21	14	267
2	238	22	12	269
3	231	29	8	273
4	236	24	8	273
5	235	25	5	276

6.1. Segmentation results

Accurate segmentation and localization of lung nodules is crucial for lung cancer detection utilizing medical imaging, such as CT scans. DSC is used as the performance metric to evaluate the effectiveness of the proposed novel segmentation method. It measures the degree of overlap between segmented regions created by the proposed 3D-VNet segmentation method, i.e., the predicted segmentation and the matching ground truth regions. The DSC value ranges from 0 to 1, with 1 denoting perfect overlap between the segmentation and ground truth and 0 denoting complete lack of overlap. The performance of the proposed 3D-VNet with that of 3D state-of-the-art segmentation architectures is summarized in Table 2. Using DSC as the performance metric, the accuracy of the proposed method is quantitatively assessed and compared to other established techniques in the field. It can be noted that the proposed segmentation architecture has performed better with a 99.3% DSC value when compared to using DSC as the evaluation method with other state-of-the-art models.

6.1.1. Analysis and discussion

Xiao et al. [45] proposed a lung nodule segmentation method using 3D-UNet and Res2Net, achieving a DSC of 95%. However, their method has limitations, including high training costs and a relatively high number of false positives. These difficulties highlight how challenging it is to do precise segmentation for tasks involving medical imaging. The authors have concluded that future advancements in 3D detection and segmentation methods will address these limitations and lead to further development in the field. Sathish et al. [47] proposed a two-stage framework for lung nodule detection in CT slices. Their method achieved an impressive DSC of 98.4% through 10-fold cross-validation, indicating an improvement over existing models. Notably, their model is computationally less expensive as it utilizes 2D CNNs without incorporating depth or volumetric information.

In contrast, this work incorporates 3D volumes of data, which come with a higher computational cost. This approach proves efficient in terms of performance, considering the complexity and richness of information obtained from volumetric data. The 3D-VNet architecture examines the complete lung scan volume in three dimensions to effectively capture the context across different slices of the lung CT images. Precise delineation of lung nodules is essential due to their shape, size, and texture variations. 3D-VNet is structured to perform complete, i.e., end-to-end segmentation. This allows it to segment lung nodules from raw volumetric scans without requiring manually constructed features or preprocessing procedures. This reduces the probability of losing essential data while preprocessing. Fig. 11 consists of (a) input CT scans from patients, (b) segmented mask, and (c) predicted mask as proposed by 3D-VNet architecture. The mask predicted by the model is similar to the segmented mask (ground truth).

6.2. Classification results

The classification training model utilizes the 3D-ResNet architecture. A combination of nodules and non-nodules was taken to ensure a balanced dataset. The non-nodules were randomly selected to achieve this balance, which may lead to varying accuracy and efficiency of the model. Due to this randomness, the model's performance may exhibit various degrees of accuracy and efficiency. This study involves the creation of multiple balanced datasets by randomly selecting non-nodules. A multi-trial technique is used to test the model's effectiveness, rather than typical cross-validation. This approach allowed us to thoroughly test the robustness and consistency of our model under varying conditions.

Case 1: Balanced dataset without augmentation

A balanced dataset was used for training in five trials without applying 3D augmentation. The average Accuracy, Sensitivity, Specificity, and FPR for five trials are 93.8%, 96.7%, 90.7%, and 9.3%, respectively. Fig. 12 presents each trial's Accuracy, Specificity, Sensitivity, and F1-score values. The corresponding confusion matrix for the first trial is displayed in Fig. 14 and the subsequent trials are displayed in Table 3.

Case 2: Balanced dataset with augmentation

Another balanced dataset was used and subjected to 3D augmentation. This set underwent training through five trials. Each trial produced distinct Accuracy, Sensitivity, Specificity, and F1-Score values. These values are depicted in a line graph in Fig. 13. The relevant confusion matrix for the initial trial is shown in Fig. 15 and the confusion matrices for the subsequent trials are shown in Table 4. These five trials for different evaluation metrics yielded different values. The average values derived for Accuracy, Sensitivity, Specificity, and FPR are 99.2%, 98.8%, 99.6%, and 0.4%, respectively.

Inference: Insufficient training data is widely recognized as a factor leading to inadequate approximation. The scarcity of data presents a common challenge in developing DL models. This lack of training data can result in inaccurate approximations of the underlying data patterns. DL models need significant representative, diverse data to perform effectively on new, untried cases. The model may need help capturing the complicated connections in the data when training data is sparse. Underfitting and overfitting are two frequent issues that might arise from limited data. When the model is excessively constrained, it fails to adequately fit the limited training dataset, while an under-constrained model tends to overfit the training data, resulting in subpar performance. An overfit model has difficulty generalizing to new data because it memorizes the training data, which includes noise and outliers. An underfit model performs poorly even on the training set and cannot recognize the underlying patterns in the data.

The accuracy of DL models relies heavily on the training data's quality, quantity, and contextual significance. Noisy or biased data might distort model outputs and predictions. However, collecting vast data, particularly images, is often impractical or unfeasible in certain cases. Collecting a big, diversified dataset can frequently be difficult or impossible. This is where data augmentation comes into play. Data augmentation involves artificially expanding the dataset by generating new data points based on existing data. As a result, there is less chance of overfitting, and the model can learn more robust features. This also helps to add diversity and variability to the training data.

Data augmentation is a type of regularization that helps the model learn more generic aspects instead of memorizing particular samples.

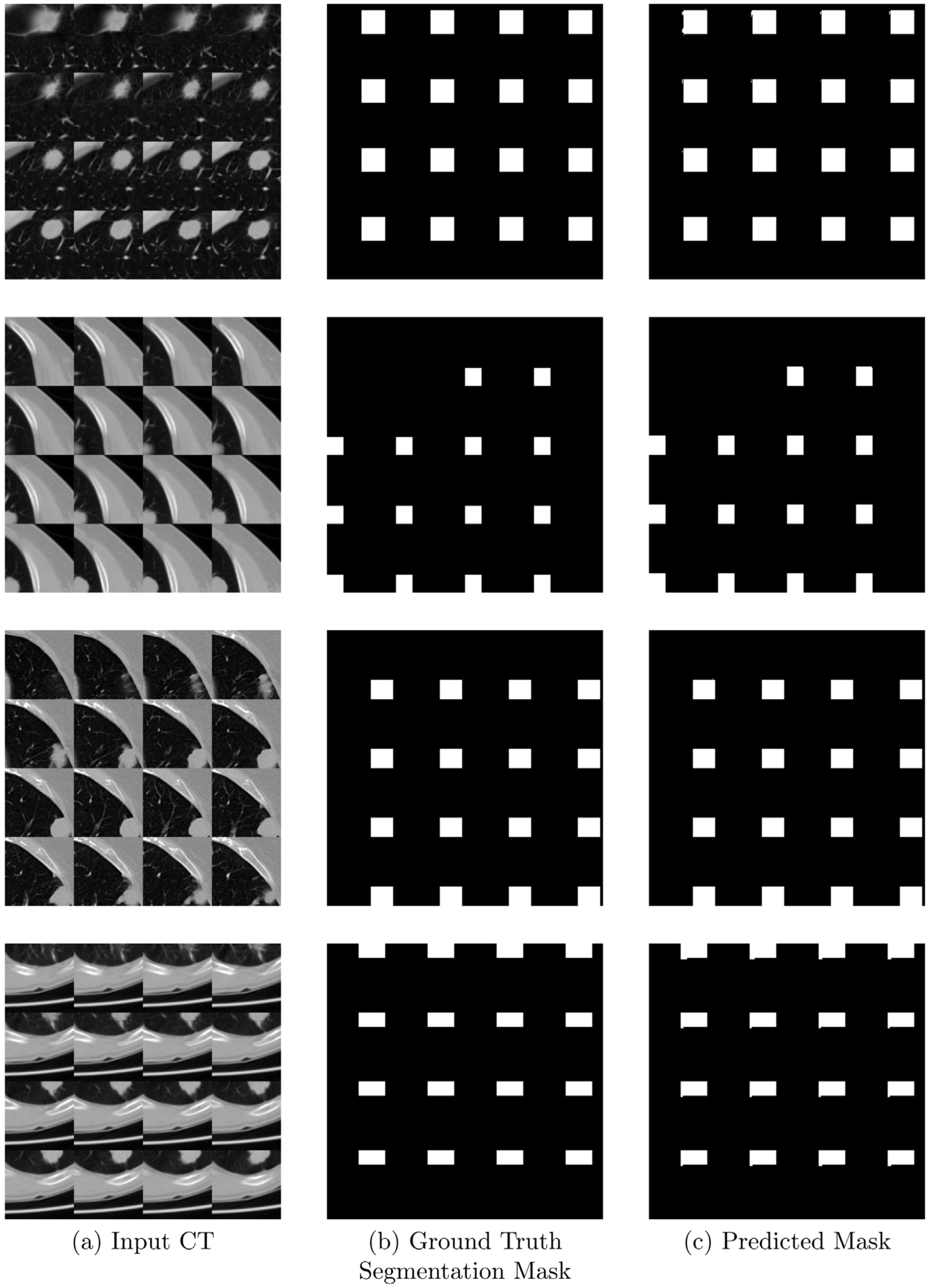


Fig. 11. Results achieved for the proposed 3D-VNet.

Regularization techniques prevent the model from incorporating noise into the data and enhance its capacity to generate precise predictions on fresh, unforeseen data. Without augmentation or regularization methods, deep neural networks are prone to learning misleading associations and memorizing intricate patterns challenging for humans to discern. This notion becomes evident in Case 1, where a balanced dataset undergoes no augmentation, remaining small. Consequently, no matter how extensively we train the model, its efficiency falls short compared to the augmented approach. In Case 2, the model receives superior training by augmenting the training dataset in each instance. As a result, Case 2 consistently delivers better results across all evaluation metrics.

Table 4
Confusion matrix for Case 2.

Trial no.	TN	FP	FN	TP
1	42 871	329	85	43 115
2	43 139	61	800	42 400
3	43 176	24	800	42 400
4	43 129	71	800	42 400
5	42 868	332	0	43 200

6.2.1. Analysis and discussion

The architecture proposed by Dodia et al. [11] for lung cancer detection, utilizing the VNet model and a combination of SqueezeNet and

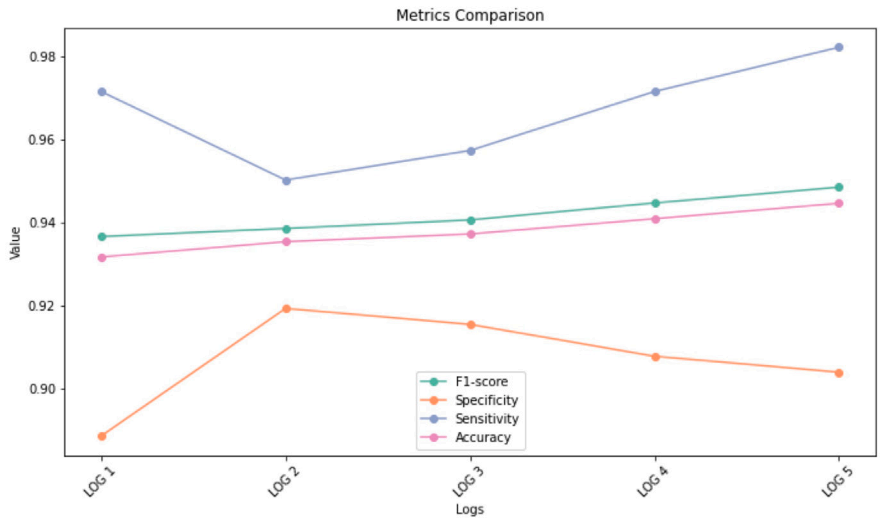


Fig. 12. Performance metrics for Case 1.

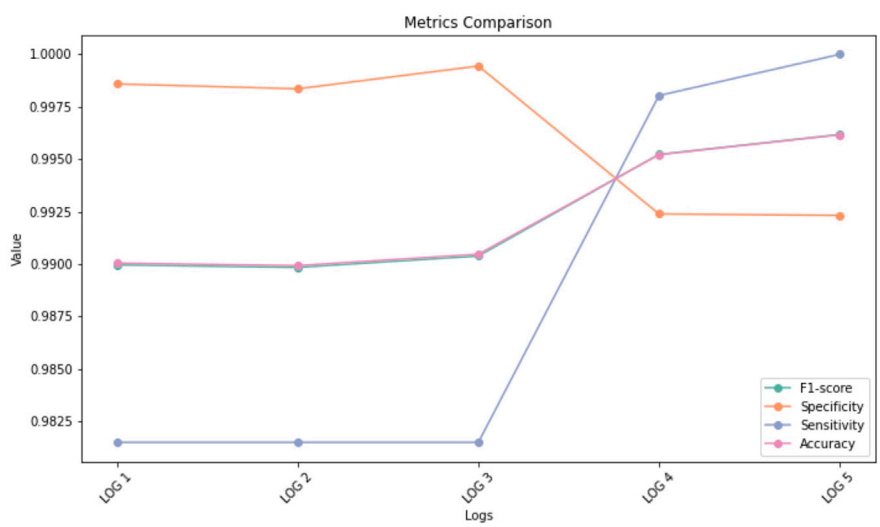


Fig. 13. Performance metrics for Case 2.

Table 5
Comparison of 3D CNN classification architectures on LUNA16.

Reference	Architecture	Accu (%)	Sens (%)	Spec (%)	FPR (%)
Ye et al. [48]	VNet model and SVM classifier	66.7	93.4	NA	NA
Liao et al. [49]	3D CNN and a leaky noisy-OR gate	81.4	NA	NA	NA
Zhou et al. [50]	VNet	84.8	96.7	99.7	NA
Bansal et al. [46]	Deep 3D segmentation network (Deep3DSCan)	88.3	87.1	89.7	1.0
Jiang et al. [51]	Ensembling 3D-Dual Path Networks (DPNs)	90.2	92.0	NA	11.1
Bharti et al. [52]	VNet and 3D ResNet	95.0	98.2	NA	NA
Dodia et al. [11]	Regularized VNet (RFR V-Net), NCNet	98.2	98.3	NA	2.3
Gong et al. [53]	3D tensor filtering by using a Correlation Feature Selection(CFS) method	NA	84.6	NA	2.8
Ours	**Proposed 3D-ResNet Model**	**99.2**	**98.8**	**99.6**	**0.4**

[1] Not Available (NA), [2] Accuracy (Accu), [3] Sensitivity (Sens) [4] Specificity (Spec), [5] False Positive Rate (FPR).

ResNet, was employed for lung nodule classification. The authors incorporated a pseudo-coloring image enhancement technique to enhance the grayscale 2D CT slice. In their VNet architecture, they implemented the regularization of receptive fields in the convolutional layers. The model achieved a sensitivity of 98.38% and an FPR of 2.3% for 3D representations. However, it was observed that the model suffered from overfitting due to using a 110-layered network with limited training data. When the training data is scarce, models tend to memorize the training examples instead of learning meaningful representations, which hampers their ability to generalize well to unseen data.

This work combines 3D-VNet with 3D-ResNet for lung cancer detection and classification. The 3D-ResNet architecture allows the network to learn more complex representations without experiencing deterioration, a typical issue observed in conventional deep networks. This depth is essential for rationalizing minute characteristics that separate malignant nodules from non-cancerous ones. 3D-ResNet utilizes global average pooling near the end of the network instead of relying heavily on fully connected layers, which are susceptible to overfitting. This decreases the overall number of parameters, directing the network towards the determinative aspects for classification. After segmentation

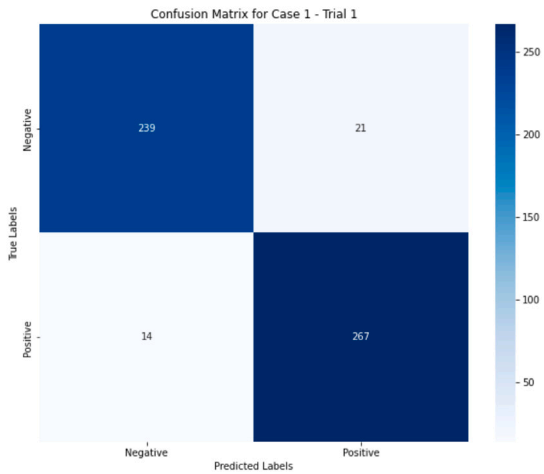


Fig. 14. Confusion Matrix for Case 1 - Trial 1.

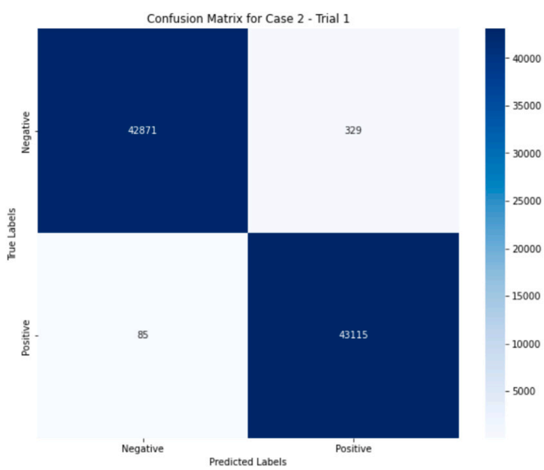


Fig. 15. Confusion Matrix for Case 2 - Trial 1.

of the lung nodules by the 3D-VNet architecture, the characteristics extracted by 3D-ResNet can be used for categorizing nodules. Based on the extracted features, the 3D-ResNet uses a softmax layer at the output to categorize nodules into cancerous and non-cancerous classes. The work aimed to generate a balanced dataset by randomly selecting non-nodules and creating an augmented balanced dataset using 3D augmentation techniques. By increasing the size of the training dataset and conducting multiple training trials, the model aimed to enhance its generalization capability. The comparison chart for 3D-ResNet with state-of-art classification models and the results are summarized in Table 5. The results of the proposed architecture averaged over multiple trials, demonstrated significant improvements over the other cutting-edge models. The average accuracy achieved was 99.2%, with a sensitivity of 98.8% and a specificity of 99.6%. FPR was drastically reduced to 0.4%. The decrease in the FPR suggests that the model successfully reduced false positive predictions.

7. Conclusion and future work

7.1. Conclusion

In the realm of medical imaging and lung cancer diagnosis, the past few years have witnessed a persistent challenge in the detection of lung nodules. Lung cancer is often first manifested as lung nodules, making early detection imperative to improve patient outcomes. The detection of lung nodules using CAD techniques presented significant challenges. CT offers high-resolution imaging, but the sheer volume of data necessitates efficient automated diagnosis. With the advent of DL algorithms, which are now at the forefront of computer-aided diagnostic techniques, these challenges are addressed quickly. CNNs, a subset of DL, have shown promise in medical image analysis. However, challenges persist, including limited annotated datasets, class imbalances, and similarities.

To address these challenges, the study proposes a novel DL architecture that employs 3D representations of CT images for segmentation and classification. A 3D-VNet architecture is introduced for precise nodule segmentation, while a 3D-ResNet architecture is developed for classification. The results achieved by this novel architecture represent substantial advancements across multiple evaluation metrics. The segmentation model achieves a DSC of 99.34% and reduces false positives to a mere 0.4% on the LUNA16 dataset. This metric reflects the model's exceptional precision in delineating and isolating lung nodules from CT scans. The classification model attains impressive performance metrics, with an accuracy of 99.2%, a sensitivity of 98.8%, and a specificity of 99.6%. The proposed DL architecture outpaces existing models and excels in lung nodule segmentation and classification. These developments can potentially increase early lung cancer detection, make more precise treatment plans possible, and ultimately improve patient care. It reaffirms the use of CAD systems to complement and enhance the expertise of medical professionals in lung cancer diagnosis.

7.2. Future work

Future research on this novel DL structure for lung nodule segmentation and classification could explore various ways to improve the model's effectiveness, broaden its scope, and incorporate it into real-world medical practice. Some of the promising directions of future research are as follows.

- The model can be trained on diverse datasets of varied types of populations, geography, or data collection organizations. This would further ensure the robustness of the model and its adaptability in generalizing when used with different imaging protocols, devices, and demographics, thus validating its performance on external datasets. Incorporating supplementary imaging techniques such as PET and MRI into CT scans can improve the precision and dependability of the model. Integrating data from various imaging sources may offer a more thorough understanding of the nodules, thus enhancing classification accuracy.
- The results could be used in future clinical trials by creating techniques to enhance the model's clarity and comprehensibility. This would aid in assessing the model's influence on diagnostic processes, which would further help evaluate the tangible advantages and pinpoint any obstacles in clinical evaluation benefiting the medical professionals and the patients. The enhancements can help grasp the rationale behind the model's predictions by showing feature importance or employing methods to emphasize specific locations in the scans that impact the model's interpretations and conclusions.
- The model can be used in long-term research to track nodule growth over time, which can aid in distinguishing between benign and malignant nodules by analyzing growth patterns. This would prove beneficial in the early detection of aggressive malignancies and in assessing the success rate of treatments.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The study did not involve any data collection. The Dataset analyzed for the current study is available at https://zenodo.org/doi/10.5281/zenodo.2595812.

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